

5. The table below gives examples of transverse and longitudinal waves.

Transverse waves	Longitudinal waves
light	sound
S waves	P waves

(a) Describe the difference between transverse and longitudinal waves.

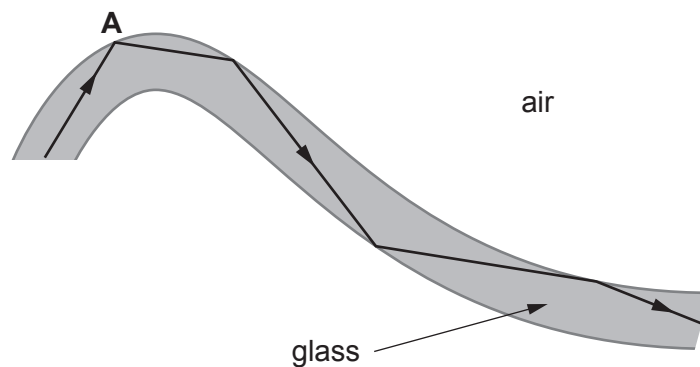
[2]

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(b) Light waves can travel through an optical fibre in the way shown in the diagram.



(i) Name the effect where light changes direction as shown at point **A**.

[1]

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(ii) Explain why light waves travel through the fibre as shown.

[2]

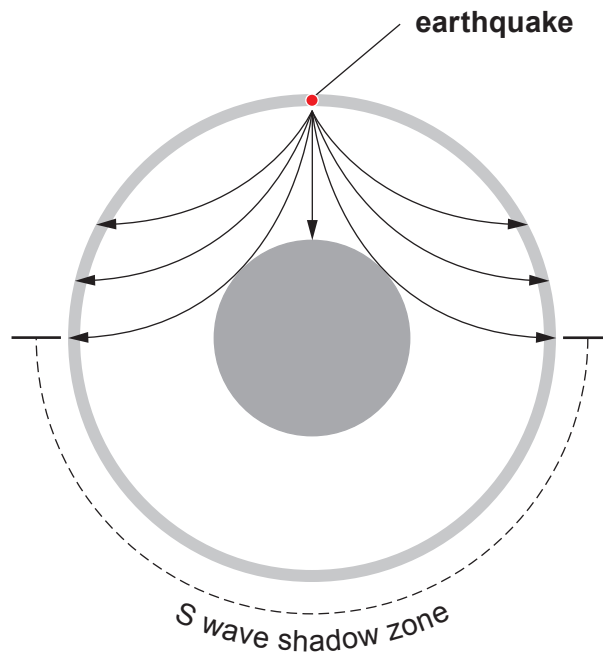
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(c) The diagram below shows an S wave shadow zone.



Explain how this led to the development of a model for the structure of the Earth. [2]

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